

EXTENDING LIFE OF YOUR SOLAR SMF BATTERY

When fresh charging fails due to successive cloudy days, you may invariably end up discharging your solar SMF battery deeply and thus drastically reducing its life. This can be avoided if the battery has automatic protection against deep discharge

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Off-grid photovoltaic (PV) solar power equipment require a battery to store the energy accumulated during sunny hours for use at night. In most applications, lead-acid batteries are preferred as these are maintenance-free and do not emit any harmful gas during the charging process.

Usually, PV solar power units use 12V or 2V sealed maintenance-free valve-regulated lead-acid (SMF VRLA) batteries specified in ampere-hour (Ah) capacity. Ideally, a 12V, 40Ah lead-acid battery will give discharge

current (called 'C') of 40A for one hour or 10A for 4 hours. However, such a battery, which is otherwise ideal for solar application, does not perform well at 1C discharge rate.

To obtain a reasonably high discharge capacity, manufacturers commonly rate these batteries at 0.1C or 10-hour discharge; that is, 4A for a 40Ah battery. Remember that even the best lead-acid battery, at this slow a discharge rate, never gives 100 per cent energy output of its rated capacity.

The life of a lead-acid battery depends on, and is directly related to, how deeply the battery is discharged. For instance, a 12V, 40Ah battery can ideally store $12 \times 40 = 480$ Wh of energy, but to avoid damaging the battery or drastically reducing its life, you should not extract all its stored energy.

When the battery is discharged, its terminal voltage starts dropping. You should stop discharging the battery when the terminal voltage reaches 10.5V minimum. The battery will certainly be damaged irreversibly if it is discharged any further.

Some batteries come with automatic protection against this. In these batteries, discharging beyond a limit is prevented unless the battery is recharged. This feature is especially useful for solar use at home since home users cannot detect whether they are discharging the battery too deeply.

When fresh charging fails due to successive cloudy days, one will invariably end up discharging one's battery deeply and, in the process, damaging it or drastically reducing its life. Therefore such deep-discharge protection must be made mandatory.

The charge-discharge cycle life of a sealed lead-acid battery is also directly related to its repetitive discharge rate. In solar applications, the battery is charged once every sunny day. So one day is one charge-discharge cycle and the battery will provide good service for:

1. 200 to 220 days (one year) with 100 per cent depth of discharge (full discharge)
2. 400 to 450 days (two years) with 50 per cent depth of discharge (partial discharge)
3. 1000 days (four years) with 30 per cent depth of discharge (shallow discharge)
4. 1500 days (six years) with 15 per cent depth of discharge (rated discharge)

There are SMF batteries available for solar use which come with integral protection to assure four to five years of minimum battery life. The protection feature automatically prevents the battery from being discharged beyond 1.7V per cell, and also does not allow it to be discharged at rates higher than 0.15C of its rating. Such batteries do not require any maintenance. ●

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